

Physics 1101W: Leg Elevator Warm Up

Sage Buesing

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1 Warm-Up questions

1.

2. See the diagram above.

We can assume that these forces are in equilibrium for both the hanging weight and the cart itself since we are trying to model a leg in traction. This is because a leg in traction should be stationary- it wouldn't really make any sense otherwise. As such, a stationary object is not accelerating and so by Newton's second law the forces on that object must be in equilibrium.

3. We shouldn't expect the normal force to equal the weight, because the cart is resting on an incline. Thus the normal force is not directly aligned with weight, and so only a portion $W \cos \theta$ of the weight is in the direction perpendicular to the slope.

4. If we put our coordinate axis with the x axis aligned with the slope, we note that for the x-direction force to be net 0, the spring force must balance with both the tension resulting from the weight of the hanging weight and the x component of the weight of the cart itself.

5. See diagram in 1

6. The equations for the forces as written on the diagrams become
For the cart:

$x :$

$$F_{NET}^x = -F_s - W_c \sin \theta + T = M_c a_c^x = 0 \quad (1)$$

$y :$

$$F_{NET}^y = N + W_c \cos \theta = M_c a_c^y = 0 \quad (2)$$

$$\implies N = W_c \cos \theta \quad (3)$$

And for the hanging mass, only the y direction has any forces. Note that the coordinate system I defined for the hanging mass is different than the

coordinate system for the cart.

$$F_{NET}^y = +T - W_h = m_h a_h^y = 0 \quad (4)$$

$$\implies T = W_h \quad (5)$$

7.

$$F_s = k\Delta x \quad (6)$$

$$W_c = M_c g \quad (7)$$

$$W - h = m_h g \quad (8)$$

2 Prediction

Assumptions: We can solve for the forces involved We are assuming that the angle of the slope is constant, and that the cart is stationary (as this is the case we are trying to model). We also assume that friction/rolling friction of the cart is negligible, as is that of the pulley. The mass of the spring and string are negligible.

We can plug in the solved forms of N and T into (1) to find

$$-F_s - W_c \sin \theta + T = 0 \quad (9)$$

$$F_s = T - W_c \sin \theta \quad (10)$$

$$k\Delta x = m_h g - M_c g \sin \theta \quad (11)$$

$$\Delta x = \frac{g(m_h - M_c \sin \theta)}{k} \quad (12)$$